MOHAMAD IYAD AL-KHIAMI

MSc, MIEAust, CPEng, NER, APEC IntPE (AUS)

Senior Instructor in Civil Engineering | PhD Candidate

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Australian University, Kuwait

LinkedIn Profile | Google Scholar | ORCID | ResearchGate

PROFESSIONAL SUMMARY

Chartered Professional Engineer (CPEng) and Senior Civil Engineering Instructor specializing in construction technology innovation, digital transformation, and occupational safety. Expertise in Industry 4.0 applications, virtual reality, artificial intelligence, and exoskeletons for construction workforce enhancement. Proven track record in curriculum development, engineering education, and bridging academic research with industry practice through innovative pedagogy and applied research.

Professional Metrics: 6+ years in industry experience | 4+ years teaching experience | 20+ peer-reviewed publications | USD 100K+ in research funding | CPEng licensed engineer | 6+ graduate supervisions

EDUCATION

PhD in Civil Engineering **2023 - 2026 (Expected)**

Aalborg University, Denmark | Department of the Built Environment

Dissertation: "Industrial Exoskeletons in Construction sectors' Workforces: Adoption Dynamics, Ergonomic Efficacy, and Productivity Gains in Mitigating Work-Related Musculoskeletal Disorders"

Supervisor: Dr. Søren Munch Lindhard

Focus: Construction Management and Informatics

Expected Graduation: August 2026

Master of Science in Civil Engineering **2015 - 2017**

Chalmers University of Technology, Sweden

Major: Design and Construction Project Management

Merit Scholarship: Excellence in Academic Performance (2016)

Thesis: The Impact of BIM on Design Phase Productivity - The West Link Case Study

Additional Coursework: System Design for Supply Chain Management and Logistics (2017)

Additional Coursework: Diploma in Nordic Built Projects - Sustainability Management and Refurbishment *Norwegian University of Science and Technology (NTNU), Norway (2017)*

Bachelor of Science in Civil Engineering **2010 – 2015**

Awards: Excellence in Academic Performance (2012, 2014)

Full Funded Scholarship for undergraduate studies

ACADEMIC APPOINTMENTS

Senior Instructor **2025 - Present**

Department of Civil Engineering, Australian University, Kuwait

Promoted from Instructor position. Teaching 70-80 students per semester across 12 undergraduate courses. Leading research initiatives in construction technology, digital innovation, and safety engineering. Supervising graduation projects and contributing to curriculum development and accreditation processes.

Instructor **2021 - 2025**

Department of Civil Engineering, Australian University, Kuwait

Delivered undergraduate courses in structural design, construction management, and engineering fundamentals. Developed new curriculum content and supervised award-winning graduation projects. Contributed to Engineers Australia accreditation achievement.

Research Assistant & Teaching Assistant **2015 - 2017**
Chalmers University of Technology, Sweden

Conducted research in construction management while pursuing graduate studies. Led revision sessions for construction management courses.

RESEARCH FUNDING

Revolutionizing Worker Well-being and Productivity in Construction **2024 - 2025**
Australian University Internal Research Grant
Amount: KWD 1,500 (USD 4,907) | **Grant #:** IRC-2023/2024-SRC-PR02
Role: Principal Investigator
Focus: Integration of Industrial Exoskeletons to Reduce Work-Related Musculoskeletal Disorders
Innovation: First comprehensive study on exoskeleton adoption in Middle Eastern construction sector

Development of a Novel Low-Energy Building **2021 - 2023**
Kuwait Foundation for the Advancement of Science (KFAS)
Amount: KWD 29,526 (USD 95,951) | **Grant #:** CN19-35EM-06
Role: Co-Investigator
Focus: Utilization of Phase Change Materials, Cool Roofs, and Solar-Assisted Air Conditioning
Impact: Sustainable building solutions for hot climate regions

RESEARCH INTERESTS

Industry 4.0/5.0 • Construction Digitalization • Occupational Health & Safety • Construction Productivity • Building Information Modeling • Virtual Reality Applications • Artificial Intelligence • Machine Learning • Construction Robotics • Ergonomics • Exoskeletons Technology • Engineering Education • Musculoskeletal Disorders • Gamification Applications

PUBLICATIONS

Journal Articles

Al-Khiami, M. I., Lindhard, S. M., & Wandahl, S. (2025). Paradox in Practice: Work Related Musculoskeletal Disorders Prevalence and Reporting Among Construction Workers in Kuwait and Denmark. Journal of Construction Engineering and Management. https://doi.org/10.1061/JCEMD4/COENG-15899

Al-Khiami, **M. I.**, Lindhard, S. M., & Wandahl, S. (2025). Prevalence of work-related musculoskeletal disorders and associated sociocultural reporting behaviors in Kuwait's construction industry. WORK: A Journal of Prevention, Assessment & Rehabilitation. https://doi.org/10.1177/10519815251325780

Al-Khiami, M. I., Jaeger, M., Soleimani, S. M., & Kazem, A. (2024). Enhancing concrete structures education: Impact of virtual reality on motivation, performance and usability for undergraduate engineering students. Journal of Computer Assisted Learning, 40(1), 306–325.

Al-Khiami, **M. I.**, Lindhard, S. M., & Wandahl, S. (2024). Integrating exoskeletons in the construction sector: A systematic review of empirical evaluation tools and future directions. Engineering, Construction and Architectural Management.

Sedaghat, A., Soleimani, S. M., **Al-Khiami**, **M. I.**, et al. (2024). Case studies on energy performance of walling materials in various regions. International Journal of Environmental Science and Technology, 21(12), 7941–7968.

Sedaghat, A., Mahdizadeh, A., Narayanan, R., Salem, H., Hussam, W. K., **Al-Khiami**, **M. I.**, Malayer, M. A., Soleimani, S. M., Sabati, M., Rasul, M., & Kamal Khan, M. M. (2023). Implementing Cool Roof and Bio-PCM in Portable Cabins to Create Low-Energy Buildings Suitable for Different Climates. Sustainability, 15(20), Article 20. https://doi.org/10.3390/su152014700

Sedaghat, A., Kalbasi, R., Narayanan, R., Mehdizadeh, A., Soleimani, S. M., Ashtian Malayer, M., **Iyad Al-Khiami**, **M.**, Salem, H., Hussam, W. K., Sabati, M., Rasul, M., & Masud K. Khan, M. (2024). Integrating solar PV systems for energy efficiency in portable cabins: A case study in Kuwait. Solar Energy, 277, 112715. https://doi.org/10.1016/j.solener.2024.112715

Al-Khiami, **M. I.**, & Jaeger, M. (2023). Safer Working at Heights: Exploring the Usability of Virtual Reality for Construction Safety Training among Blue-Collar Workers in Kuwait. *Safety*, *9*(3). https://doi.org/10.3390/safety9030063

Al-Khiami, **M. I.**, Jaeger, M., & Soleimani, S. M. (2023). The Integration of Digital Techniques in Engineering Education: A Case Study to Evaluate Student's Motivation and Performance. *International Journal of Information and Education Technology*, *13*(5), 844–848. https://doi.org/10.18178/ijiet.2023.13.5.1877

Sedaghat, A., Salem, H., Hussam, W. K., et al. (2023). Exploring energy-efficient building solutions in hot regions: A study on bio-phase change materials and cool roof coatings. *Journal of Building Engineering*, *76*, 107258. https://doi.org/10.1016/j.jobe.2023.107258

Conference Publications

Al-Khiami, M. I., Lindhard, S. M., & Wandahl, S. (2025). Sustainable Workforce Health: Exoskeleton Effects on Posture and Performance in Repetitive Lifting for Labor-Intensive Industries and Construction. *CIB Conferences*, *1*(1). https://doi.org/10.7771/3067-4883.1660

Al-Khiami, **M. I.**, Soleimani, S. M., & Jaeger, M. (2025). A Comparative Study of VR and 2D Plans in Concrete Structure Evaluation for Engineering Education: Does Perceived Usability Affect Performance? 2025 IEEE Global Engineering Education Conference (EDUCON), 1–7.

Al-khiami, **M. I.**, Lindhard, S. M., & Wandahl, S. (2024). Human-centric innovation in construction: 19th International Strategic Management Conference. https://isma.info/ismc2024/?p=conference-programme

Al-Khiami, **M. I.**, & ElHadad, M. M. (2024). Enhancing Construction Site Safety Using AI: The Development of a Custom YOLOV8 Model for PPE Compliance Detection. 2024 European Conference on Computing in Construction, Chania, Crete, Greece.

Al-Khiami, **M. I.**, Lindhard, S. M., & Wandahl, S. (2024). Charting the Exoskeleton Industry: A Comprehensive Insight into Dynamics and Trends. IOP Conference Series: Earth and Environmental Science, 1389(1), 012014. https://doi.org/10.1088/1755-1315/1389/1/012014

Al-Khiami, **M. I.**, Jaeger, M., & Soleimani, S. M. (2023). A Gamified Approach to Concrete Structure's Interpretation using Virtual Reality: A Study of Performance and Simulation Sickness. 2023 IEEE Global Engineering Education Conference (EDUCON), 1–6.

Al-Khiami, **M. I.**, Jaeger, M., & Soleimani, S. M. (2023). The Use of Disruptive Technologies in the Construction Industry: A Case Study to Compare 2D and VR Methods of Concrete Design Interpretation. *Proceedings*, *85*(1), Article 1. https://doi.org/10.3390/proceedings2023085001

Sedaghat, A., Al-Masri, A., **Al-Khiami**, **M. I.**, & Malayer, M. A. (2023). Practical Considerations for Phase Change Materials in Kuwaiti Building Envelopes: A Data-driven Approach. 2023 Computer Applications & Technological Solutions (CATS), 1–6. https://doi.org/10.1109/CATS58046.2023.10424376

Sedaghat, A., Salem, H., Hussam, W. K., Mahdizadeh, A., **Al-Khiami**, **M. I.**, & Malayer, M. A. (2023). Experimental Study on Cool Roof Methods for Two Identical Portable Cabins in the State of Kuwait. ICMAME 2023 Conference Proceedings. International Conference on Mechanical, Automotive and Mechatronics Engineering, Dubai, UAE. https://doi.org/10.53375/icmame.2023.267

Book Chapters

Sedaghat, A., Nazififard, M., & **Al-Khiami**, **M. I.** (2025). Dynamic IoT Determination of Overall Heat Transfer Coefficient in a Portable Cabin in Kuwait. In G. Rey, J.-Y. Tigli, & E. Franquet (Eds.), Internet of Things (pp. 279–290). Springer Nature Switzerland.

Soleimani, S. M., **Al-Khiami**, **M. I.**, Sedaghat, A., & Malayer, M. A. (2024). Harnessing Solar Energy for Efficient Cooling in Hot Weather Regions. In X. Wang (Ed.), Future Directions in Energy Engineering (pp. 69–77). Springer Nature Switzerland.

Preprints & Other Publications

Al-Khiami, **M. I.**, & Jaeger, M. (2023). Leveraging ChatGPT in Android App Development: A Case Study on Supporting Novice Developers in Creating Successful Apps. *Preprints*. https://doi.org/10.20944/preprints202307.0660.v1

Al-Khiami, **M. I.**, & Zangana, A. (2017). The Impact of BIM on Design Phase Productivity: The West Link Case Study. *Master's Thesis*, Chalmers University of Technology. https://hdl.handle.net/20.500.12380/251912

CURRICULUM DEVELOPMENT & ACADEMIC LEADERSHIP

Civil Engineering Program Accreditation

Contributed to curriculum development that successfully achieved Engineers Australia accreditation for the Civil Engineering program at Australian University, Kuwait. Participated in curriculum design, course mapping, learning outcome assessment, and accreditation documentation to meet international engineering education standards and professional engineering requirements.

New Course Development

• Developed and implemented Building Information Modeling (BIM) course - first of its kind in the program
• Created comprehensive course content integrating industry-standard BIM practices with academic theory
• Designed practical laboratories and hands-on learning experiences for emerging construction technologies

Course Content Innovation

• Developed new slides and comprehensive course content for Design of Concrete Structures
• Created Revit Design course materials integrating 3D modeling with structural design principles
• Redesigned Civil Engineering Materials course content with modern testing methods and sustainable materials

• Integrated digital technologies and virtual reality applications into traditional engineering courses
• Updated course materials to reflect current industry practices and emerging technologies

TEACHING EXPERIENCE

Current Courses Taught

Design of Concrete Structures • Design of Steel Structures • Civil Engineering Materials • Basic Construction • Construction Management • Statics • Strength of Materials • Engineering Skills • Project Based Learning • Engineering Ethics • Capstone Design • Engineering Drawings

Teaching Load: 70-80 students per semester | **Total Students Taught:** 500+ students
Teaching Innovation: Integrated VR technology, AI applications, and digital tools in engineering education

Student Supervision (Bachelor's Level Graduation Projects)

• Investigating Exosuit Efficacy: A Controlled Study of Physiological Impacts in Work-Related MSD Prevention (2024-2025) - Second Place Winner
• Ergonomics in Construction: Mitigating Work-Related MSD Using Exosuits (2024-2025)
• Automated PPE Compliance Detection: Using Artificial Intelligence for Construction Workers (2022-2023) - Second Place Winner
• Machine Learning for Safety Equipment: Detecting Defects in Safety Harness and Lanyard in Construction (2022-2023)
• Assessing the Re-Usability of Plywood Post Concrete Casting using AI (2022-2023)
• The use of Virtual Reality Training for Safety and Hazard Detection Among Blue-Collar Labor in Construction (2022-2023)

Supervision Success Rate: 100% completion rate | 33% award-winning projects

PROFESSIONAL EXPERIENCE

Civil Engineer, Design & Implementation 2019 - 2021

Dar Al-Dabbous Engineering & Consulting Bureau, Kuwait

Responsible for design review, construction planning, scheduling, and site inspection of various infrastructure projects.

Site Engineer 2017 - 2019

Tarmeem Trading & Contracting Company, Kuwait

Site engineer for structural building execution in the Kuwaiti Refinery project, managing construction operations and quality control.

Research Assistant 2016 - 2017

Trafikverket (Swedish Transportation Agency), R&D Department

Evaluated the integration of Building Information Modeling (BIM) in transportation infrastructure projects as part of master's thesis research.

Project Scheduler 2015 - 2017

Dar Al-Dabbous Engineering & Consulting Bureau, Kuwait

Prepared detailed time schedules for construction projects based on Bill of Quantities analysis. Developed project timelines, resource allocation plans, and critical path analysis for multiple concurrent projects. Coordinated with project managers and field teams for schedule optimization.

Civil Engineer 2014 - 2015

Spectrum Trading & Contracting Company, Kuwait

Site engineer for execution of various commercial building projects in Kuwait. Managed construction supervision, quality assurance, and coordination with design teams. Gained foundational experience in construction project management and site operations.

PROFESSIONAL SERVICE

Editorial & Review Activities

• Conference Reviewer: IEEE EDUCON 2025 Global Engineering Education Conference
• Journal Reviewer: Journal of Construction Engineering and Management (ASCE)
• Peer Review Service: Multiple international journals in construction engineering and education technology
• Review Expertise: Construction safety, digital technologies, BIM applications, engineering education

Committee Memberships

• Member of the Technology and Virtual Reality Committee
• Member of the Validation and Moderation Committee
• Member of the Accreditation Committee
• Member of the Auditing & Quality Management Committee
• Member of the Departmental Disciplinary Committee

PROFESSIONAL CERTIFICATIONS & MEMBERSHIPS

Professional Engineer Certifications

• **Chartered Professional Engineer (CPEng)** - Engineers Australia (EA), May 2023 - Present
• **International Professional Engineer (IntPE AUS)** - Engineers Australia (EA)
• **National Engineer Register (NER)** - Australia
Specializations: Civil Engineering, Construction Engineering, Project Management

Specialized Technical Certifications (2023-2024)

• **Ergonomics in Construction** - NYC Construction Safety Training LLC (May 2024)
• **Building AI Powered Chatbots without Programming** - IBM (March 2024)
• **Generative AI Essentials** - Coursera (March 2024)
• **Data Science for Construction, Architecture and Engineering** - EdX Verified Certificate (March 2024)
• **Programming for Everybody (Getting Started with Python)** - EdX Verified Certificate (Feb 2024)
• **Python Data Structures** - EdX Verified Certificate (Feb 2024)
• **Systematic Literature Review: A Practical Guide** – Udemy (July 2023)
• **YOLOv7 YOLOv8 Deep Learning Computer Vision** - Udemy (Apr 2023)
• **Primavera P6 Industry Based Training** – Udemy (Nov 2020)
• **Managing Projects with Microsoft Project** – Microsoft (Feb 2016)

Professional Development

• **Project Management Professional (PMP) Contact Hours:** 35 hours completed
• **Continuing Professional Development:** Active participation in engineering education and construction technology training

Professional Memberships

• **Member of Engineers Australia (EA)** - 2023 to Present
• **Member of Kuwait Society of Engineering (KSE**) - 2015 to Present

TECHNICAL SKILLS

Programming & Data Science: Python, Data Analysis, Machine Learning, Statistical Analysis

Civil Engineering Software: AutoCAD, Revit, SAP2000, Primavera P6, Microsoft Project

Digital Technologies: Unity3D, Virtual Reality (VR), Building Information Modeling (BIM), IoT Systems

Office & Productivity: Microsoft Office Suite, Project Management Tools, Academic Writing

LANGUAGES & AWARDS

Languages
Arabic: Native Speaker
English: Fluent (Academic and Professional)

Communication Skills
• International conference presentations • Academic writing and publication • Cross-cultural collaboration • Technical report writing • Grant proposal development • Student mentoring and supervision

Selected Awards & Honors

• Merit Scholarship for Excellence in Academic Performance - Chalmers University (2016)
• Excellence in Academic Performance - Kuwait University (2012, 2014)
• Full Funded Scholarship - Kuwait University (2010)
• Supervision of Award-Winning Graduation Projects (First Place, 2 Second Place Winners)